

Quantum-Enhanced Cloud-Conditioned Diffusion Models for Cloud Removal in Sentinel Imagery: A Proof of Concept Study

Introduction

Cloud cover removal has been a fundamental challenge in the Earth Observation community that has usually been overcome by utilizing time-series imagery along with traditional cloud removal algorithms such as FMASK and S2Cloudless [2]. Deep learning based approaches have started holding merit in recent years especially since advances in diffusion models have demonstrated superior performance over GAN-based approaches, offering stable training and high-fidelity reconstruction. Furthermore, quantum machine learning has shown promising results in Earth Observation, with hybrid quantum-classical architectures achieving significant improvements in image synthesis tasks—including 64% lower Fréchet Inception Distance (FID) and 76% lower Kernel Inception Distance (KID) compared to classical baselines on the EuroSAT dataset [1].

Despite these advances, the application of quantum-enhanced diffusion models to cloud removal in Sentinel imagery remains unexplored. This gap presents an opportunity to investigate whether quantum computing techniques can improve the reconstruction fidelity and spectral consistency of cloud-conditioned diffusion models.

Background Research or Context

Cloud removal in Sentinel-2 imagery using neural networks has been an active field of research and development since its inception, though adoption in commercial workflows has only been rather recent. This is mainly due to significant improvements in cloud removal and consequent image generation capabilities, initially offered by Generative Adversarial Networks (GAN) but later superseded by Diffusion models, or more specifically Cloud-conditioned variants that achieve state-of-the-art fidelity and stability. While in a GAN architecture, a generator and discriminator try and compete to create fake clean images and spot fakes respectively - it is a rather cumbersome process wherein mode collapse and vanishing gradients are common, causing artifacts and overall sub-par image synthesis when compared to Diffusion models [7]. In a cloud-conditioned diffusion model, the core mechanism is similar to a diffusion model, wherein gradually noise is added (forward pass) and then the networks learns to remove it (backward pass) but with an additional step to condition on a cloudy image at each step [1].

As for the quantum aspect, quantum machine learning has demonstrated substantial promise for Earth Observation tasks, particularly through hybrid quantum-classical architectures that integrate quantum circuits into classical neural networks. The Quantvolutional Conditioned U-Net (QCU-Net), for instance, strategically replaces classical convolutional layers with quantvolutional layers—quantum convolutional filters that process local image patches via angle encoding into qubits, followed by variational quantum circuits (VQCs) with entangling gates—and achieves significant improvements over purely classical diffusion baselines [4]. Unlike GANs' instability or classical diffusion's parameter-intensive feature extraction, these quantum enhancements leverage superposition and entanglement to capture complex spatial-spectral correlations more expressively in high-dimensional Hilbert space, often performing better in data-scarce scenarios typical of EO applications [4,6]. However, such quantum diffusion models have thus far focused on generative tasks like class-conditional image synthesis, leaving their potential for reconstructive cloud removal.

Proposed Project

This proof-of-concept study is aimed to develop on pre-existing research in cloud-conditioned diffusion models by incorporating suitable *quantvolutional* layers to evaluate quantum-enhanced variants of cloud-conditioned diffusion models for Sentinel-2 cloud removal. The project will:

1. **Baseline Implementation:** Establish a classical cloud-conditioned diffusion baseline on the current SOTA (DiffCR) [3] trained on a subset of the SEN12MS-CR dataset [5].
2. **Quantum Enhancement:** Integrate quantvolutional layers and variational quantum circuits (VQCs) into the diffusion U-Net architecture, adapting quantum modules from recent quantum EO diffusion work to the cloud-removal conditioning mechanism.
3. **Experimental Validation:** Compare DiffCR and quantum-enhanced variant on standard metrics (MAE, LPIPS, etc) across varying cloud coverage conditions.
4. **Analysis:** Assess computational trade-offs, performance across the Sentinel-2 bands and indices, and generalization to unseen regions.

The scope is deliberately constrained to 64×64 (maybe even 32×32 or smaller depending on training times) image chips and quantum simulators to ensure feasibility within the time-frame while establishing a foundation for future large-scale studies.

Possible research questions are:

1. To what extent do quantum-enhanced cloud-conditioned diffusion models improve the reconstruction fidelity of cloud-conditioned diffusion models for Sentinel-2 imagery compared to purely classical baselines?
2. Can quantum-enhanced cloud-conditioned diffusion models achieve comparable cloud removal quality to classical diffusion with significantly less training data, as demonstrated in quantum generative tasks?

Timeline

Part-Time

February 28 - April 2: Further Literature Review and Initial DiffCR Setup

Full-Time

May 12 - May 19: Training and Baseline

May 20 - June 5: Dataset preparation

June 6 - June 20: Model Adaption

June 21 - July 5: Training and Initial Evaluation

July 6 - July 20: Hyperparameter Tuning and Final Evaluations

July 21 - August 6: Writing the Report

August 7 - August 15: Polishing code and report and Preparation for Viva

*These are tentative estimates and sub-parts might get shorter or longer depending on results but project will nevertheless finish by August 15, 2026.

References

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